Use Case Diagram: Sample Solution
University Library System

Identify the actors

1. Student: Searches for, borrows, and returns books.

2. Librarian: Manages book records and checks overdue fines.

3. Timer (System Scheduler): Triggers automatic overdue reminders.

Identify the use cases

No.	Use Case
1	Search for Books
2	Borrow Books
3	Return Books
4	Manage Book Records
5	Check Overdue Fines
6	Send Overdue Reminders
7	Check Book Availability
8	Assess Overdue Fine

Relationships

Association (solid line): Connects actors to the use cases they perform.

<<include>> relationship: Borrow Books includes Check Book Availability because availability must be checked before borrowing.

<<extend>> relationship: Assess Overdue Fine extends Return Books, but only when a returned book is overdue. The arrow points from Assess Overdue Fine toward Return Books.

System boundary

The rectangle is labelled University Library System. All use cases are inside the rectangle, while the actors are outside.

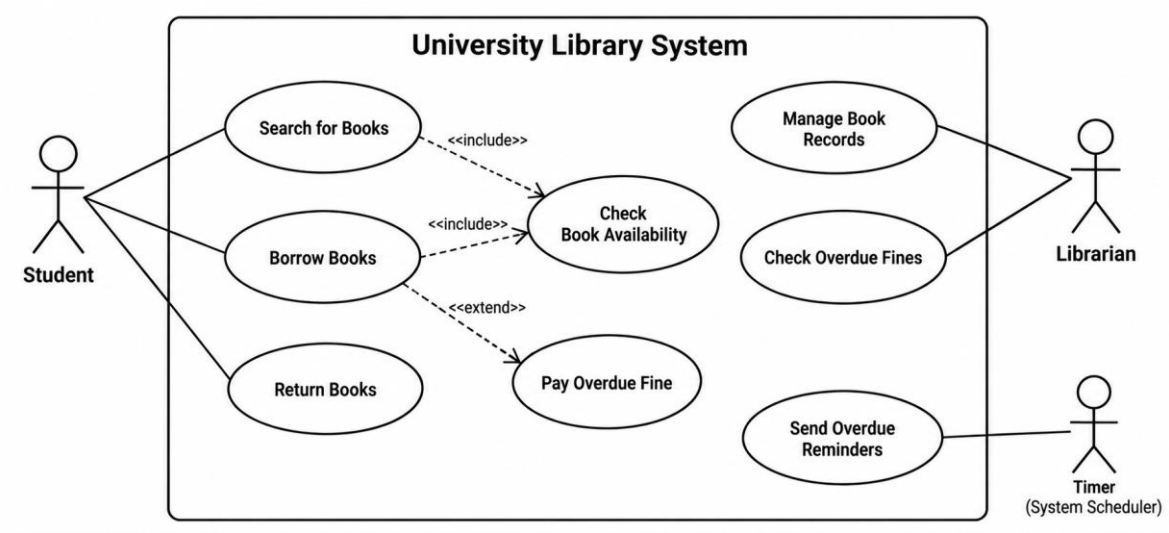


Figure 1: Use Case diagram for the University Library System